

Experiment – 6

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Simulation of UI Design Lifecycle using RAD Model

INTRODUCTION

This experiment focuses on simulating the **UI design lifecycle using the RAD (Rapid Application Development) model**. The goal is to quickly design and develop an **interactive interface**. The application created is **CosmoPDF**, a PDF toolkit with a space-themed UI.

RAD MODEL PHASES APPLIED

1. Requirement Planning Phase

- Identified user needs:
 - Merge, split, compress, and convert PDF files
- Decided features:
 - File upload
 - Tool selection
 - Output download

2. User Design Phase

- Created UI layout with:
 - **Galaxy/space theme** background
 - Tool cards (Merge, Split, Compress, etc.)
 - Search bar and category filters
- Focus:
 - Attractive design
 - Easy navigation
 - Clear visual hierarchy

3. Construction Phase

- Developed interactive interface:
 - File upload (drag & drop / browse)
 - File selection display
 - Options like filename, page order, bookmarks
 - “Merge Files” action button
- Implemented smooth transitions between screens

4. Testing & Deployment Phase

- Tested with sample PDF files
- Verified:
 - File selection
 - Merging process
 - Output generation
- Final screen shows:
 - **“Mission Complete”** confirmation
 - Download option

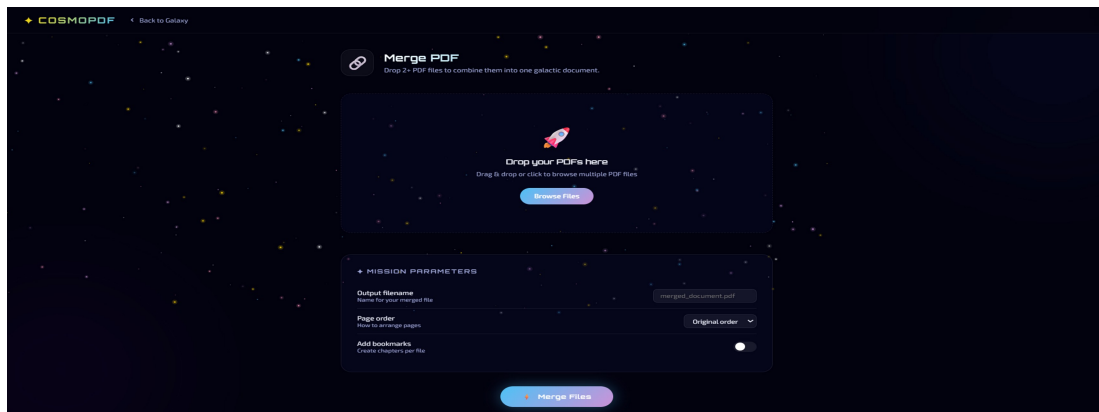
INTERFACE FLOW

1. Home Screen



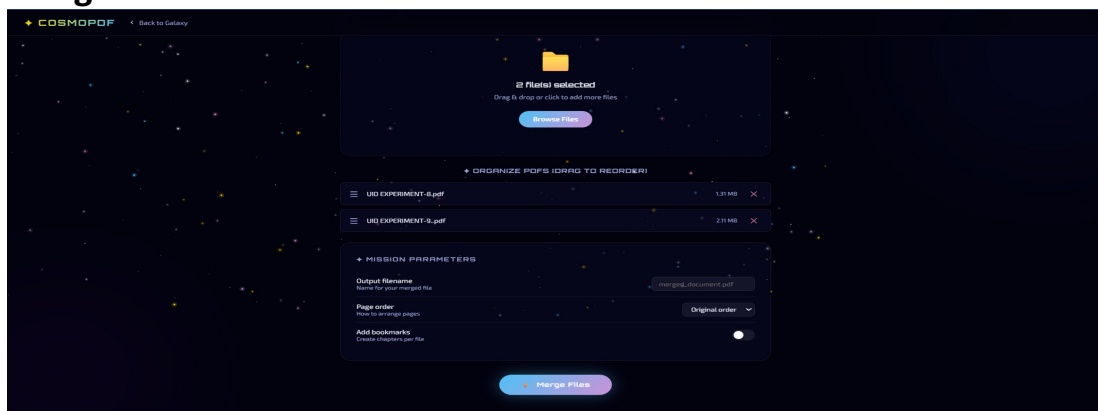
- Displays all PDF tools
- Search and category filters
- User selects **Merge PDF**

2. Upload Screen



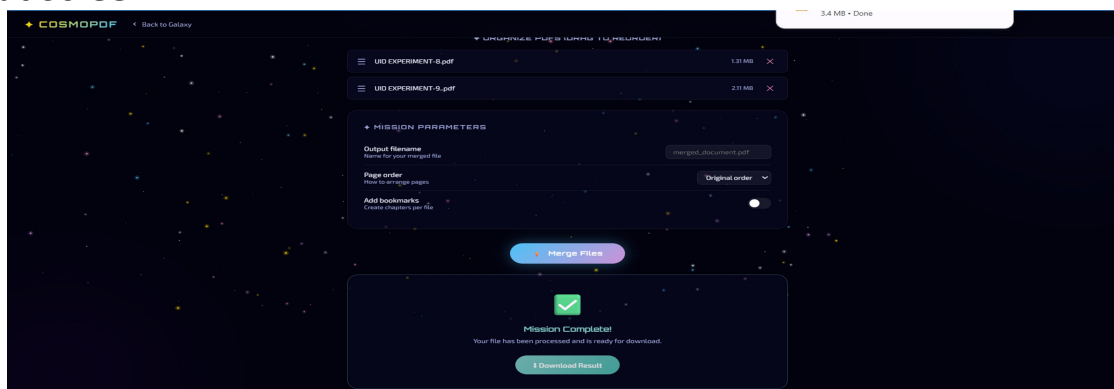
- User uploads PDF files
- Drag & drop or browse option

3. File Management Screen



- Shows selected files
- Allows reordering and removal
- Additional settings (filename, bookmarks)

4. Output Screen



- Displays success message
- Provides download button

CONCLUSION

This experiment demonstrates how the **RAD model enables rapid UI development** with continuous improvements. The CosmoPDF interface provides a smooth, interactive experience with clear workflow from **tool selection to final output**.